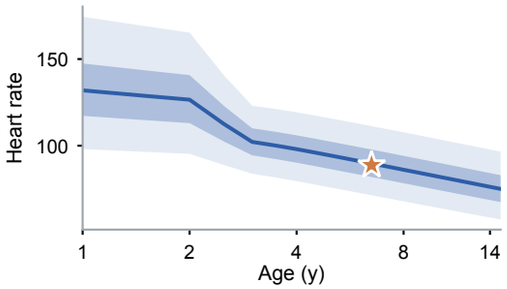
A Simulator pipeline: one synthetic ECG for a chosen age, sex and rhythm

1 Age profile

Age 1–14 y | **Boy or girl**

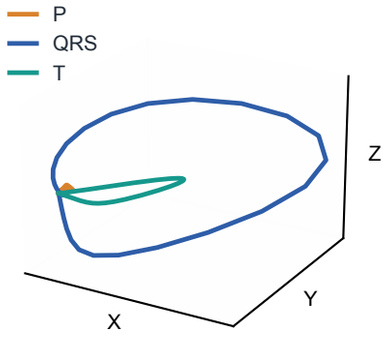


Draw HR · PR · P · QRS · QT · RMSSD

Drawn together, so they fit each other

QT follows heart rate (age-band QT–RR)

2 Heart-vector beat (VCG)

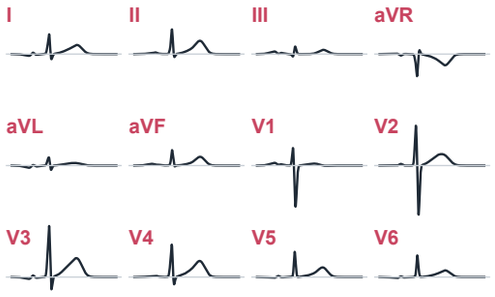


8 smooth waves: 2 P · 4 QRS · 2 T

Timing from step 1 placed exactly

Shape and size drawn by age (783 children)

3 12-lead projection



Each child's own lead directions

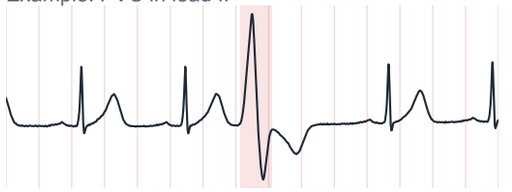
Atrial field (P) + ventricular field (QRS, T)

Limb leads by Einthoven / Goldberger

4 Rhythm and recording

Sinus	normal rhythm for this age
Sinus tachycardia	rate above age p98
Sinus bradycardia	rate below age p2
PAC	early beat, narrow QRS
PVC	early, wide, no P, full pause

Example: PVC in lead II



Noise: baseline wander · broadband · 50 Hz

Output: synthetic 12-lead ECG · 10 s · 500 Hz · exact rhythm label and beat-by-beat annotations

B Data used

ZZU pECG · ages 1–14
799 strict-normal children (development set)

Published references
Rijnbeek 2001 · PHN 2018 (toddler heart rate)

Inverse Dower matrix
Vondrák et al. 2022 (12 leads ↔ X, Y, Z)

C Checks before use (measured blind)

+2 / +8 ms QRS / QT error, rebuilding 200 real children	**84–95%** normal ECGs inside age p2–p98 (real: 96%)	**85–99.5%** ECGs that keep the intended rhythm
0.88 AUROC, real vs synthetic (0.5 = identical)	**2.1 vs 1.8 y** age-recovery error: synthetic vs real	**T axis** weakest: 77% inside p2–p98 (too wide)

D Role in the main study

A **Age-conditioned diffusion**
learns from real ECGs + age only

B **Same model + this simulator**
profile timing as extra input;
pretrained on simulated ECGs

Panel C checks are reused to judge both A and B